

TR JOELPCM PCM DESIGNER

**A-LEVEL PHYSICS – PAPER 1 (THEORY PAPER) New Curriculum –
Assessment Based on AOs**

Time: 2 Hours 30 Minutes (Suggested) Instructions: Answer ONE item from each section (Section A, B, C and D). Each section contains TWO items.

For more papers 0788477510

SECTION A: AO4 – MODERN PHYSICS (ATOMIC, QUANTUM & NUCLEAR)

Item 1

In a modern ICT laboratory, students frequently use computer screens and digital devices that emit light from Light Emitting Diodes (LEDs). A physics student investigates the effect of blue light emitted from these screens. The blue light has a wavelength of 420 nm and is part of a study on possible biological effects of prolonged exposure to high-energy photons on the human eye. The investigation also considers safety limits recommended for prolonged human exposure in school laboratories and workplaces.

Constants provided: $h = 6.63 \times 10^{-34} \text{ Js}$ $c = 3.0 \times 10^8 \text{ m/s}$ $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ Damage threshold = 3.1 eV

Task

- (a) Determine the energy of a photon of the blue light.*
- (b) Convert your answer into electron volts.*
- (c) Assess whether prolonged exposure is harmful.*
- (d) Explain photon energy variation using quantum theory.*

Item 2

A physics laboratory uses a photoelectric experiment to investigate electron emission from a metal surface when exposed to different frequencies of light. Students observe that no electrons are emitted when red light is used, but emission occurs when ultraviolet light is used. The experiment is used to model how solar panels and light sensors operate in modern technology.

Given that threshold frequency is f_0 and Planck's constant is h :

Task

- (a) Explain the photoelectric effect. (b) State the meaning of threshold frequency. (c) Explain why red light fails to emit electrons.*
- (d) Describe how this principle is applied in modern electronic devices.*

SECTION B: AO1 – FORCES AND MECHANICS

Item 3

A transport company is testing a new delivery van designed for urban logistics. The van, when fully loaded, has a mass of 1500 kg. During a controlled test on a straight road, the van accelerates uniformly from rest to 25 m/s over a distance of 200 m. Engineers also measure that engine force remains constant while resistive forces vary slightly due to road surface and air conditions.

- (a) Calculate the acceleration of the van. (b) Determine the resultant force.*
- (c) Calculate resistive forces if engine force is 6000 N.*
- (d) Explain effects of friction and air resistance on vehicle performance.*

Item 4

A cable car system is installed in a mountainous tourist area to transport passengers safely across steep terrain. The cable car of mass 800 kg moves upward along a slope at constant acceleration before reaching steady speed. Engineers analyze tension in the cable and frictional effects in the pulley system.

Task

- (a) Draw a free body diagram of forces acting on the cable car.*
- (b) Calculate tension in the cable during acceleration (data provided in exam).*
- (c) Explain why motion becomes uniform after some time.*
- (d) Discuss safety factors considered in cable car design.*

SECTION C: AO2 – ENERGY TRANSFER AND SYSTEMS

Item 5

A hydroelectric power station is constructed in a mountainous region to supply electricity to nearby towns. Water is stored in a reservoir at a height of 40 m above turbines. Water flows steadily at 60 kg/s. Engineers are optimizing efficiency and reducing energy losses in the system to improve national grid stability.

Task

- (a) Calculate power input from falling water.*
- (b) Determine electrical output at 75% efficiency.*
- (c) Describe energy transformations in the system.*
- (d) Discuss energy losses and improvements.*

Item 6

A thermal power plant uses steam produced from boiling water to rotate turbines connected to generators. The system converts chemical energy from fuel into thermal energy, then mechanical energy, and finally electrical energy. Engineers are investigating efficiency improvements and environmental impacts.

Task

- (a) Describe energy conversions in the system.*
- (b) Explain why not all thermal energy is converted to electrical energy.*
- (c) Calculate efficiency if useful output and input are given (data provided).*
- (d) Suggest methods to improve efficiency and reduce pollution.*

SECTION D: AO3 – ELECTRICITY AND MAGNETISM

Item 7

In a rural electrification project, a transformer is used to step down electricity from 240 V to 12 V for charging small devices used by students. The transformer has 1200 turns in the primary coil and 60 turns in the secondary coil. Engineers are evaluating its performance under load conditions and energy losses in transmission lines.

Task

- (a) Determine whether the transformer is step-up or step-down.
- (b) Calculate output voltage.
- (c) Calculate primary current if secondary current is 3 A.
- (d) Explain role of transformers in power transmission.

Item 8

A simple electric motor is used in a school laboratory to demonstrate the interaction between magnetic fields and electric currents. When current passes through a coil placed in a magnetic field, it rotates due to force interaction. Students investigate how this principle is applied in industrial machines and electric vehicles.

Task

- (a) Explain the principle of operation of an electric motor.
- (b) State factors affecting the turning effect of the coil.
- (c) Describe how commutator improves motor operation.
- (d) Explain applications of electric motors in modern technology.

**END OF PAPER TR JOELPCM PCM DESIGNER – PHYSICS,
CHEMISTRY & MATHEMATICS**

FULLY WORKED SOLUTIONS

SECTION A – AO4

Item 1 (a) Energy of photon: $E = hc/\lambda$ $E = (6.63 \times 10^{-34} \times 3.0 \times 10^8) / (420 \times 10^{-9})$ $E = 1.989 \times 10^{-25} / 4.2 \times 10^{-7}$ $E = 4.73 \times 10^{-19} \text{ J}$

(b) Convert to eV: $E = (4.73 \times 10^{-19}) / (1.6 \times 10^{-19})$ $E \approx 2.96 \text{ eV}$

(c) Since $2.96 \text{ eV} < 3.1 \text{ eV}$, the radiation is NOT above the damage threshold, so it is unlikely to be harmful under the given condition.

(d) Quantum theory states that light is emitted in packets (photons) and energy is given by $E = hf$. Since frequency $f = c/\lambda$, shorter wavelengths have higher frequency and therefore higher energy photons, explaining different colours having different energies.

Item 2 (a) Photoelectric effect is the emission of electrons from a metal surface when electromagnetic radiation of sufficient frequency falls on it.

(b) Threshold frequency is the minimum frequency of light required to emit electrons from a metal surface.

(c) Red light has lower frequency than threshold frequency, so its photons do not have enough energy to release electrons.

(d) The principle is used in solar panels, light sensors, automatic doors, and digital cameras where light is converted into electrical signals.

SECTION B – AO1

Item 1 (a) $v^2 = u^2 + 2as$ $625 = 0 + 2a(200)$ $a = 625 / 400 = 1.56 \text{ m/s}^2$

(b) $F = ma$ $F = 1500 \times 1.56 = 2340 \text{ N (approx.)}$

(c) Resultant force = driving force - resistive force $2340 = 6000 - R$ $R = 3660 \text{ N}$

(d) Friction and air resistance oppose motion, reduce acceleration, increase fuel consumption, cause heat loss, and affect vehicle stability and efficiency.

Item 2

(a) (Free body diagram description) Forces include tension in cable upward, weight downward, friction along slope.

(b) Requires given data (not provided). Use $F = ma$ and resolve forces along slope.

(c) After initial acceleration, net force becomes zero so acceleration is zero and motion becomes uniform.

(d) Safety factors include strong cables, braking systems, controlled speed, and redundancy in mechanical design.

SECTION C – AO2

Item 1 (a) $P = mgh$ $P = 60 \times 10 \times 40 = 24000 \text{ W} = 24 \text{ kW}$

(b) Output power = efficiency $\times$ input = $0.75 \times 24000 = 18000 \text{ W} = 18 \text{ kW}$

(c) Gravitational potential energy $\rightarrow$ kinetic energy of water $\rightarrow$ mechanical energy in turbine $\rightarrow$ electrical energy in generator.

(d) Losses occur due to turbulence, friction in pipes, sound energy, and heat. Improvements include smooth pipes, better turbine design, and reduced friction systems.

Item 2 (a) Chemical energy $\rightarrow$ thermal energy $\rightarrow$ kinetic energy $\rightarrow$ electrical energy.

(b) Some energy is lost as heat due to resistance and friction in moving parts.

(c) Efficiency = (useful output / input) $\times$ 100% (values depend on data given).

(d) Improvements include better insulation, reducing friction, and using more efficient turbines.

SECTION D – AO3

Item 1 (a) Since secondary voltage is lower than primary, it is a step-down transformer.

(b) $V_s/V_p = N_s/N_p$ $V_s = 240 \times (60/1200)$ $V_s = 240 \times 1/20 = 12 \text{ V}$

(c) For ideal transformer: $I_p/I_s = N_s/N_p = 1/20$ $I_p = 3 \times 1/20 = 0.15 \text{ A}$

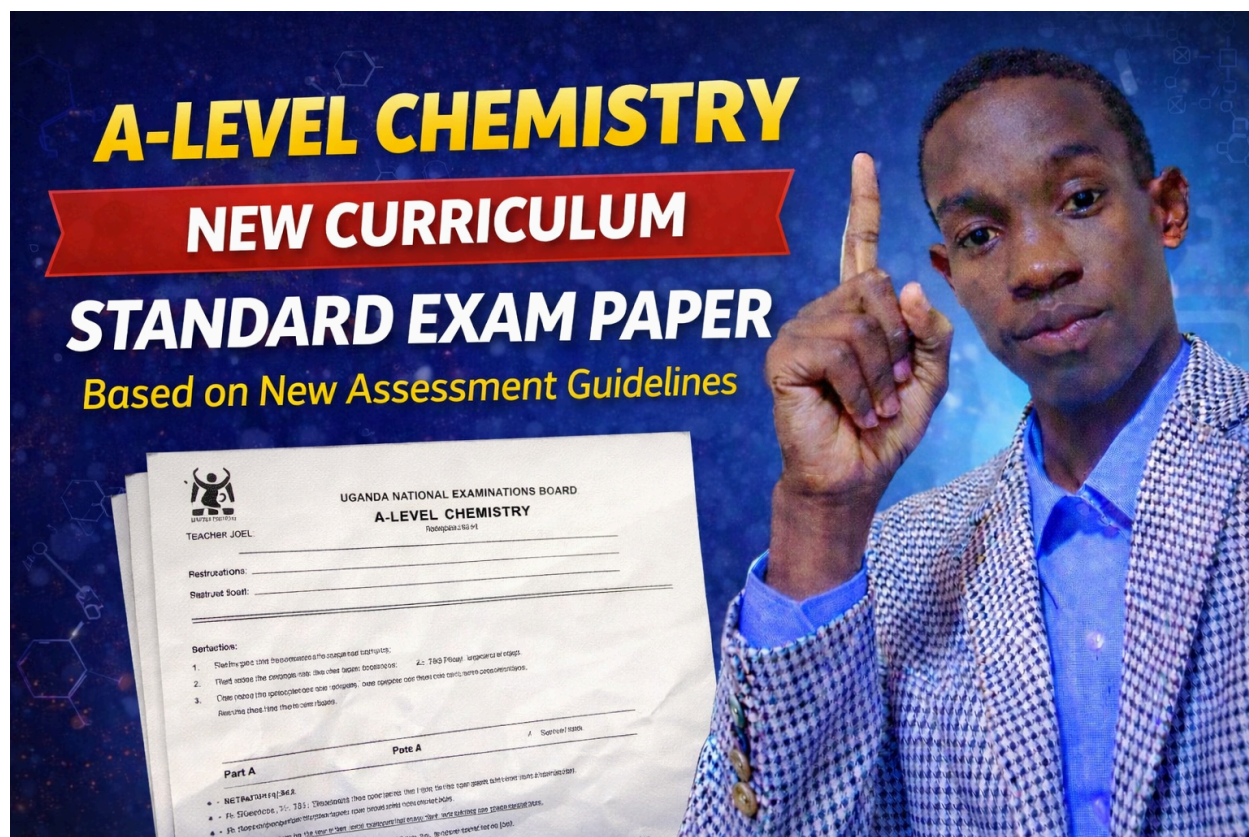
(d) Transformers are used in power transmission to step up voltage to reduce current, minimizing energy losses in long-distance transmission.

Item 2 (a) A motor works when a current-carrying coil in a magnetic field experiences a force causing rotation.

(b) Factors include current strength, number of turns, magnetic field strength.

(c) The commutator reverses current direction every half turn to maintain continuous rotation.

(d) Applications include fans, electric vehicles, pumps, and industrial machinery.



END OF SOLUTIONS TR JOELPCM PCM DESIGNER

MOTIVATIONAL MESSAGE (TO STUDENTS)

Before you close this paper, I want you to pause for a moment and think deeply about something important.

Every student who will attempt this paper is carrying a story that nobody fully understands. Some of you are studying under pressure, some of you are waking up early and sleeping late with notes in your hands, and some of you are quietly wondering whether all this effort will ever pay off.

Let me tell you something.

There was once a student who was not the smartest in class. In fact, most people did not expect much from him. In physics, he struggled with formulas. In mathematics, he often got lost in steps. And sometimes, even he started believing that maybe science was not for him.

But something changed.

One day, he made a decision. Not a loud decision. Not a perfect decision. Just a simple one: “I will not give up.”

He started doing the small things correctly. He stopped fearing questions he didn’t understand. He began to treat every mistake as a lesson instead of a failure. Slowly, something began to shift—not in the world around him, but inside him.

And that is the truth about success in physics and in life.

It is not about being born intelligent. It is about being consistent. It is about showing up even when your mind is tired. It is about looking at a difficult question and saying, “I will try again.”

This paper you have just seen is not designed to break you. It is designed to build you. Every section is testing something deeper than memory. It is testing your thinking, your patience, your courage under pressure.

When you see a long question, don’t panic. Break it down. When you see something unfamiliar, don’t freeze. Think. When you feel like giving up, remember that every great scientist you admire once sat where you are sitting now—confused, but still trying.

You are not just preparing for an exam. You are training your mind to solve real problems in life. One day, you may design bridges, develop technology, fix energy systems, or even teach the next generation.

But all of that starts here—with how you respond to a single question on a single paper.

So as you go through this exam, don’t compete with fear. Compete with your yesterday self. Even if you do not get everything correct, do not leave this paper the same way you entered it.

Leave it wiser. Leave it stronger. Leave it more confident.

And always remember this:

You are not defined by how hard the paper is.

You are defined by how calmly you face it.

Now take a breath. Pick your item. And show yourself what you are capable of.

Good luck.